Crop Alignment: A Structural-Null Test for Closed-Loop Labels in Segmentation Benchmarks

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Abstract

Many segmentation benchmarks are labelled by algorithms, not by people. When the labelling rule reads the same pixels the model reads, the benchmark closes a loop: a label stops being a property of the ground and becomes a function of the window it was computed in, and a model can score well by reproducing the labeller instead of the phenomenon. We give a test for this, calibrate it, and apply it.

The test asks, on pixels where overlapping crops disagree, how much more often a model predicts the label of the crop it is currently reading than the label a different crop assigns the same physical pixel. On any nonempty disagreement set, we prove this quantity is zero for a predictor whose output does not depend on the crop. The null is structural, so it carries no sampling error and needs no control arm to estimate. Run on a per-pixel threshold, which is crop-invariant by construction, our estimator returns $|\kappa| < 1.4 \times 10^{-15}$. The scope is deliberately narrow: the test has no support when crops never disagree and cannot diagnose crop-invariant pseudo-label shortcuts or shortcuts unrelated to crop geometry.

A structural null shows the test reads zero in a known negative control. It does not show the test is right when crop dependence is present, so we build a labeller whose crop dependence we set with a dial. In the released event-level summaries, crop alignment rises from $+0.0569$ to $+0.2476$ across that dial and is ordered correctly with it in 10 of 11 held-out events. Over the same sweep, accuracy against expert labels falls by 0.1056 mIoU ($t = -4.27$), and κ tracks that loss at fixed dial settings.

On a published sea-ice benchmark whose labeller we recover exactly, the 16 folds with committed machine-readable evidence give crop alignment $+0.1230$ for the benchmark labels and $+0.0157$ for a matched control: a paired difference of $+0.1073$ ($t = 12.88$, positive in all 16 folds). We release the test as a single file; it reproduces the committed aggregate to floating-point precision. The release does not contain the raw predictions or checkpoints needed to reproduce training end to end, and we state that evidence boundary explicitly.

1. Introduction

Auto-labelling made large segmentation benchmarks possible. It also made some of them circular, because the label is a deterministic function of the image the model receives. One measurable form of that circularity occurs when the labeller is run on crops and its answer changes with the crop. That failure is widely acknowledged and rarely quantified, in part because there has been no estimand with a value known in advance under the null that can be computed on a benchmark you did not build.

This paper supplies one. We define crop alignment, prove its null, calibrate it against a labeller we control, measure what the thing it detects costs against human annotation, and release an implementation that runs on other people's data. We apply it to two benchmarks, and Sections 8 and 9 report what it finds there.

Foundation-model masks are increasingly used as ground truth [14]. When those masks are generated crop by crop, they can inherit the property tested here; scene-level pseudo-labels can still close the input–label loop but fall outside our diagnostic if they are crop-invariant. The adjacent literatures in Section 2 diagnose related failures after the fact. What is scarce is a controlled measurement of this crop-dependent case.

Terms. A benchmark is closed-loop when its labels are computed from the model's own input. Crop alignment (κ) is defined in Section 3 and audits only the crop-dependent subset of such benchmarks. The artefact premium is defined in Section 9, where it is also estimated.

2. Related work

Closed-loop labelling sits between five literatures. The distinction this paper turns on is the one between shortcut learning and label noise.

Shortcut learning. Documented cases include hospital-specific tokens in chest radiographs [28], background rather than animal in wildlife recognition [1], and the general account of shortcut learning as a property of gradient descent rather than of any one dataset [7, 16]. In each case the cue was identified after a model had already been trained and seen to generalise badly. Crop alignment is a forward test with a value known under the null, so it can be run before anyone has been surprised, and it names the cue in advance.

Annotation artefacts. The closest analogue is not in vision. Hypothesis-only baselines on natural language inference showed that labels were partly predictable from properties of the annotation process, and that leaderboard progress had been measuring that predictability [8, 22]. The failure is the same. The difference is that a crowdworker's habits cannot be recovered, whereas an algorithmic labeller can be. We invert one labeller to its constant in 17 of 17 folds and recover another labeller's threshold and its preprocessing kernel to within 0.25 dB on 10 of 10 events. Exact recovery lets us build the artefact with a dial instead of only detecting it.

Label noise, and why this is not that. The standard frame for imperfect labels is noise, with a large literature on robust losses, noise-transition estimation, and the observation that many test labels in standard benchmarks are simply wrong [18, 19, 27]. Closed-loop labels are the opposite of noise. Noise is unpredictable from the input, so a model cannot learn it. A crop-dependent label is deterministic in the model's own input, so a model learns it exactly. That is why closed-loop labels inflate measured performance where noise depresses it. Treating closed-loop labels as noise prescribes the wrong remedy, because robustness to them makes a model worse on the benchmark and better on the truth.

Weak and programmatic supervision. Labelling functions and foundation-model pseudo-labels are the practice that produces closed-loop labels at scale [23]. We measure one failure mode of that practice, and the failure is about where the labelling function reads from. A rule applied once per scene is fine. The same rule applied per crop is not.

Benchmark statistics. A separate line argues that benchmark comparisons are routinely reported without the variance or the power to support them, that seed and data-order variation can exceed the differences being claimed [3, 24], and that many published effects are underpowered [4, 5]. Our required-sample-size framing in Section 9 applies that argument to a benchmark whose independent unit is the acquisition and not the patch, and follows the standard practice of fixing a margin in advance [15].

Spatial validation. In remote sensing and ecology, splitting spatially autocorrelated data at random inflates measured skill [21, 25], and treating correlated observations as independent replicates has a name and a long history [11]. Our protocol repairs are the standard remedies applied to a benchmark that used none of them. What we add is not the remedies but a measurement of what skipping them costs, stated as a required sample size.

3. Crop alignment

3.1. Definition

A labeller produces, for each crop c and each pixel p inside it, a label $L_c(p)$. When it reads only the pixels inside the crop, two crops covering the same source pixel can assign it different labels. A predictor f maps a pixel and the crop it is read in to a class, written $f(p \mid c)$.

For a source pixel p, let $C(p)$ be the set of crops covering it and $m(p) = |C(p)|$. The artefact set is

$$\mathcal{A} = \{\, p : \exists\, c, c' \in C(p) \text{ with } L_c(p) \neq L_{c'}(p) \,\}. \quad (1)$$

Off $\mathcal{A}$ the labeller is locally single-valued and there is nothing to detect. A pixel covered by one crop can never enter $\mathcal{A}$, so $m(p) \geq 2$ throughout.

For $p \in \mathcal{A}$ and $c \in C(p)$, with $\mathbf{1}[\cdot]$ the indicator, define

$$\mathrm{own}(p, c) = \mathbf{1}[f(p \mid c) = L_c(p)]\,, \quad (2)$$

$$\mathrm{oth}(p, c) = \frac{1}{m(p) - 1} \sum_{c' \neq c} \mathbf{1}[f(p \mid c) = L_{c'}(p)]\,, \quad (3)$$

where here and below sums over c or c' run over $C(p)$. Then

$$\kappa = \mathbb{E}\big[\, \mathrm{own}(p, c) - \mathrm{oth}(p, c) \,\big], \quad (4)$$

when $\mathcal{A}$ is nonempty, with the expectation taken uniformly over instances (p, c) with $p \in \mathcal{A}$, or under any weighting that is exchangeable across the crops covering a given pixel. Equation (2) asks whether the prediction matches the label of the crop being read. Equation (3) asks how often it matches the label of a different crop covering the same pixel, averaged so that pixels with more coverage are not overweighted.

3.2. The null is structural

Proposition 1. Suppose $\mathcal{A}$ is nonempty. Let f be crop-invariant on $\mathcal{A}$, meaning $f(p \mid c) = f(p)$ for every $p \in \mathcal{A}$ and every $c \in C(p)$. Then $\kappa = 0$ exactly.

Proof. Fix $p \in \mathcal{A}$, write $m = m(p)$ and $y = f(p)$, which by hypothesis does not depend on the crop. For each class k let $n_k = |\{c \in C(p) : L_c(p) = k\}|$, so that $\sum_k n_k = m$. Summing Eq. (2) over the crops covering p,

$$\sum_c \mathrm{own}(p, c) = \sum_c \mathbf{1}[y = L_c(p)] = n_y. \tag{5}$$

Summing Eq. (3) and exchanging the order of summation,

$$\begin{aligned} \sum_c \mathrm{oth}(p, c) &= \frac{1}{m-1} \sum_c \sum_{c' \neq c} \mathbf{1}[y = L_{c'}(p)] \\ &= \frac{1}{m-1} \sum_{c'} \mathbf{1}[y = L_{c'}(p)] \cdot (m-1) \\ &= n_y, \end{aligned} \tag{6}$$

because each c' is counted once for every c other than itself, that is $m - 1$ times, and the normaliser cancels it.

So Eq. (5) and Eq. (6) agree at every $p \in \mathcal{A}$. The average in Eq. (4) over all N instances is

$$\kappa = \frac{1}{N} \sum_{p \in \mathcal{A}} \sum_c \big[\mathrm{own}(p, c) - \mathrm{oth}(p, c)\big] = 0, \tag{7}$$

since the inner sum vanishes for each p, whatever its coverage. □

Nothing in the argument uses the number of classes, the label distribution, the coverage $m(p)$, the positive size of $\mathcal{A}$, or any property of the imagery. There is no sampling error on this null, and no control arm is needed to estimate it. If $\mathcal{A}$ is empty, however, κ has no observations and is not defined; emptiness is not evidence that the labels are free of other shortcuts.

Corollary 1. $\kappa \neq 0$ certifies that f is crop-dependent on $\mathcal{A}$.

Attributing a positive κ to a model having learned the labeller requires the matched control of Section 3.3, because a network trained on crop-invariant labels also reads positive.

3.3. Why the measured control is not zero

A convolutional network reading crops is not a crop-invariant predictor. Zero padding at the crop border, receptive fields truncated at the window edge, and any normalisation computed over the crop all make $f(p \mid c)$ depend on c near boundaries. We measure this directly as Ω, the probability that two crops covering the same pixel yield different predictions. It is 0.032 for sea-ice models trained on repaired labels and 0.081 for flood models at the zero setting of the dial in Section 5.

The empirical control therefore sits a little above zero, at +0.0157 and +0.0569 respectively. That value is a resolution floor, not a failure of Proposition 1: the proposition describes a crop-invariant predictor, and a network is not one. Every result below is the contrast between a closed-loop arm and its matched control, never raw κ.

4. Verifying the estimator, and bounding its scope

4.1. A case with a known answer

Proposition 1 supplies something rarer than a null: an input whose correct output is known before looking, and therefore a test of the code rather than of the data. A per-pixel threshold on chip-level smoothed VH reads one pixel value and cannot see a window, so it is crop-invariant literally, not approximately. The estimator must return zero.

Measured on four held-out flood events, on the same artefact set as every other arm and between 9.5 and 41 million instances per event, $|\kappa|$ is 6.8×10^{-16}, 1.3×10^{-15}, 1.0×10^{-15} and 8.5×10^{-16}, with Ω exactly zero in all four. The two probabilities in Eq. (4) agree to every printed digit, 0.5977 against 0.5977 for the first event. What remains is accumulated floating-point rounding.

Any error in the $(m-1)$ normaliser, the own-versus-other bookkeeping, or the artefact-set mask would surface here as a nonzero number. Anywhere else it surfaces as a plausible one somewhere it could not be caught.

4.2. κ is relative to the crop grid

κ is measured on pixels where covering crops disagree, and how densely crops are sampled is a choice we made. We tested whether the number survives that choice by thinning the evaluation grid from stride 32 to stride 64

grid	crops/pixel	artefact set	κ contrast
stride 32	16	24.71%	+0.1907
stride 64	4	16.31%	+0.3260

Table 1. Crop alignment measured on the same models and labels through two evaluation grids, the sparser one a strict subset of the denser.

on identical models and labels. The sparser grid is a strict subset, so nothing was recomputed and nothing retrained.

It moves, by 1.71× (Tab. 1). The sparser grid reads a larger alignment, because the artefact set shrinks to the pixels the crops disagree about most strongly and the second term of Eq. (4) averages over fewer and more distant crops. What survives is the ranking. The two geometries order the eleven events at $r = +0.984$.

The scope of the instrument is therefore narrower than it first appears. κ compares arms evaluated on the same crop geometry. Its absolute magnitude is not comparable across papers that tile differently, and a κ reported without its crop size and stride cannot be interpreted. Every comparison in this paper is made within a fixed grid.

5. Calibrating the test

The sea-ice benchmark cannot establish sensitivity, because the crop-dependence of that pipeline is the unknown being estimated. So we built a labeller with a dial. Sen1Floods11 [2] ships an Otsu-on-VH [20] label set computed once per chip, whose preprocessing we recover in Section 8.1. Holding that smoothing at the chip level and recomputing only the threshold inside each crop reproduces the sea-ice mechanism with one scalar per crop as the entire source of crop-dependence:

$$\tau_c(\alpha) = (1-\alpha)\,\tau_{\text{chip}} + \alpha\,\tau_{\text{crop}}. \tag{8}$$

At $\alpha = 0$ the labels are crop-invariant by construction. At $\alpha = 1$ the labeller is fully closed-loop. The measured spread of per-crop thresholds within a chip is 6.13 dB, so the dial has real range.

The artefact set is fixed at the $\alpha = 1$ labels and every arm is scored on those same pixels. Without that, the artefact set would shrink as α fell and vanish at $\alpha = 0$, leaving the control with no pixels to be null on. The sea-ice control works the same way, evaluating a scene-trained model against the original labels.

Tab. 2 and Fig. 1 give the result. The contrast between the ends is +0.1907 (sd 0.0212, $t = 29.84$), positive in 11 of 11 events, sign test $p = 0.00098$. Per-event rank correlation with α averages +0.991 and is perfect

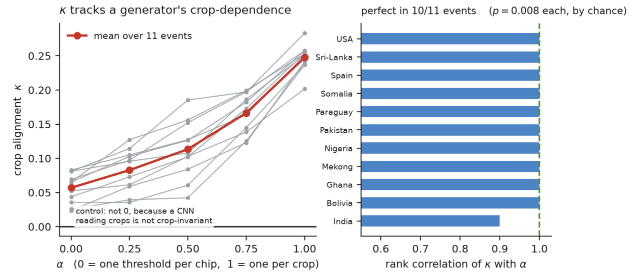

Figure 1. κ against the dial of Eq. (8). Grey lines are held-out events, red is the mean. Right: rank correlation of κ with α per event.

α	0.00	0.25	0.50	0.75	1.00
κ	+0.0569	+0.0828	+0.1134	+0.1664	**+0.2476**
Ω	0.0807	0.0841	0.0897	0.0977	0.1135

Table 2. Crop alignment and model crop-dependence against the dial of Eq. (8), averaged over the eleven held-out events.

in 10 of 11 events. The exception is a 0.0002 inversion at the floor between $\alpha = 0$ and $\alpha = 0.25$.

Positivity on closed-loop labels alone could be accidental. Reproducing the ordering across five settings, within each event, is harder to get by accident: a single event ordering five levels correctly by chance has $p = 1/120$.

Scope. Sen1Floods11 as published computes one threshold per chip, so the closed-loop artefact is not present in that benchmark. The imagery and the labeller are real and the crop-dependence is ours. This is a calibration, not a second sighting in the wild, and nothing here bears on that dataset's published labels.

6. What closed-loop labels cost

6.1. Against human annotation

The 55 models from the dial sweep were trained, validated and tested entirely inside their own labeller, and had never been scored against the expert labels the dataset ships. Scoring them against the human required no new training.

We score by mosaic: one decision per source pixel, taken by majority vote of the crops covering it, compared once against the expert label. Patch scoring would count each pixel about sixteen times and would hand the crop-dependent arms exactly the reward this paper says patch scoring hands them, which would manufacture the result rather than measure it.

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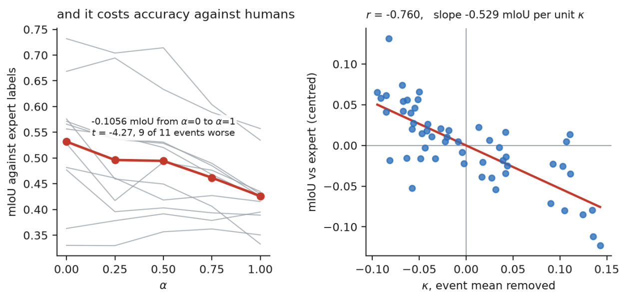

Figure 2. Left: mIoU against expert labels falls as the labeller becomes more crop-dependent. Right: κ against that mIoU with each event's mean removed, so the slope cannot come from events differing in difficulty.

α	0.00	0.25	0.50	0.75	1.00
mIoU vs expert	0.5318	0.4962	0.4944	0.4618	**0.4262**

Table 3. Accuracy against expert labels over the dial sweep of Eq. (8), mosaic scored.

α	0.00	0.25	0.50	0.75	1.00	mean
r	-0.614	-0.698	-0.542	-0.567	-0.534	$\mathbf{-0.591}$

Table 4. Correlation between κ and mIoU against expert labels, computed across the eleven events separately at each dial setting.

Accuracy against the expert labels falls monotonically with the dial (Tab. 3). The cost of full crop-dependence is -0.1056 mIoU (sd 0.0821, $t = -4.27$), worse in 9 of 11 events. Pooled across the sweep with each event's mean removed, κ tracks that loss at $r = -0.760$, a slope of -0.529 mIoU per unit κ ($t = -8.50$), shown in the right panel of Fig. 2. Two events improve, by $+0.0318$ and $+0.0203$, both among the lowest-mIoU events in the set. Absolute values are lower than this dataset's whole-chip numbers because these are 128×128 crop-trained models with less context; the quantity in use is the comparison across the dial, not the level.

6.2. Is κ only a proxy for the dial?

The dial moves κ and the damage together, so a correlation between them is equally consistent with κ being a bystander. We therefore hold α fixed and ask whether κ still separates events once the dial cannot vary.

All five settings give a negative correlation (Tab. 4). Two are individually significant at $n = 11$, the strongest at $p = 0.017$, and the other three fall between $p = 0.069$ and $p = 0.091$. The five arms are not independent, since the dial nests them by construction and they share the same eleven events, so what the table shows is consistency across the sweep and not five separate tests. At any fixed level of crop-dependence, the events where the model aligned more with the crop labeller did worse against the human.

We do not claim κ is the mechanism by which the damage happens. That would need a mediation design this experiment does not have. We claim it is an indicator that survives removing the confound.

	ice	flood-chip	flood-crop
input	RGB+10	VV,VH	VV,VH
crop/stride	128/–	512/–	128/32
LR	1e-4	3e-4	3e-4
loss	focal	CE	CE
batch	128	8	32
cap/pat.	12/5;60/15;120/15	40/10	20/6
held out	acquisition	event	event
folds	17 (16 κ)	11	11
seeds	42,7,123	42,7,123	42

Table 5. Configuration of the three experimental arms. Crops are square and sizes are in pixels. CE denotes cross-entropy.

7. Experimental setup

The sea-ice benchmark contains 169,051 patches over 39 tiles and 17 acquisitions; the flood benchmark has 446 chips over 11 events. The crop sweep draws 24 crops per chip per epoch and evaluates κ on all 169 positions of a 13×13 grid. The ten co-registered ICESat-2 photon features [17] form the second model branch in Fig. 3; that branch is held fixed across every arm.

Splits are leave-one-acquisition-out or leave-one-event-out, the unit sharing a labeller constant. Validation uses 15% of the remaining units and never the held-out unit. A tile-disjoint arm also forbids tiles from crossing train and test. The 12-epoch sea-ice arm uses shorter patience, so it is not a pure cap change. Reported runs used two RTX A6000 cards capped at 300 W: sea-ice folds took 41–67 minutes at 60 epochs and 49–149 at 120; flood crop runs took about four minutes.

8. The instrument in the wild

8.1. Two labellers, recovered exactly

Sea ice. The labeller of [12], scaled with cloud and shadow removal in [13], applies HSV thresholds to each RGB patch after background subtraction. The thresholds admit hue 0 to 185 while OpenCV hue ranges 0 to 179, so hue and saturation constrain nothing and only the value channel acts. A two-parameter grid search recovers the water constant in 17 of 17 folds. Otsu and

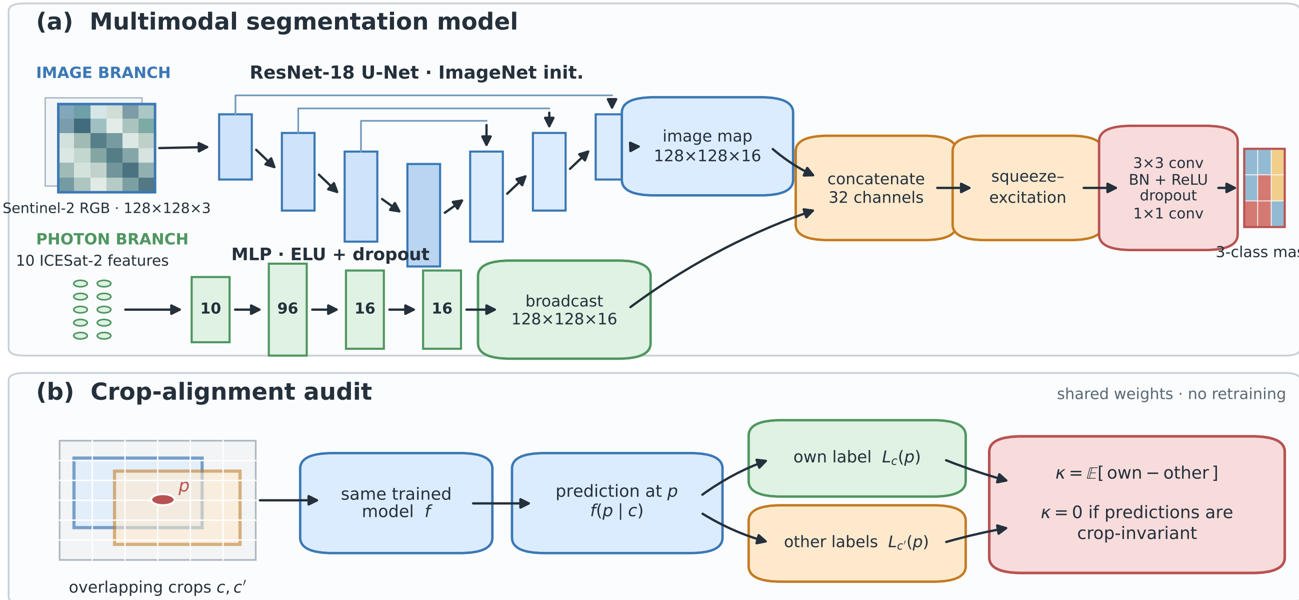

Figure 3. Code-faithful model and audit. An ImageNet-initialized [6] ResNet-18 [9] U-Net [26] encodes each image; an MLP encodes the photon features; and squeeze–excitation [10] fuses them. The audit sends overlapping crops through shared weights and compares agreement with the current crop's label against other crops' labels for the same source pixel.

two min-max normalisations are recomputed on each 128×128 patch, so a pixel's label depends on the framing of its crop. Regenerating once per scene makes the labels exactly crop-invariant: crop unanimity is 1.0000 on all 17 folds for the repaired scheme against 0.9678 for the original.

Floods. Sen1Floods11 ships an expert label and an Otsu label for the same 446 chips, and unlike the sea-ice case it publishes the constant, so the recovery can be checked from outside. Recovery from the raw band gives 94.9% pixel agreement but sits 1.19 dB low on all ten events with a published value. A bias of one sign everywhere is a signature, not noise. Sweeping the smoothing radius, the bias crosses zero at a 9×9 focal mean, where 10 of 10 events land within 0.25 dB at 99.05% agreement (Fig. 4). The constant moves 5.57 dB across events, so no global threshold can express it and a model seeing the whole chip can infer which event it is in and adapt.

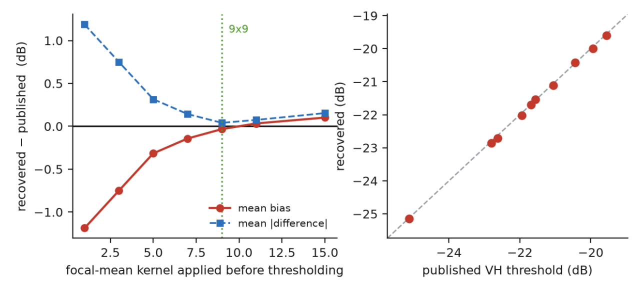

Figure 4. Recovering the flood labeller. Left: bias and mean absolute difference against the published thresholds as the smoothing kernel is swept. Right: recovered against published threshold, one point per event.

8.2. Crop alignment on the sea-ice benchmark

Sea-ice crops overlap about 33 times, so a source pixel is seen inside many crops, and under the per-crop scheme those crops can disagree. For the 16 folds with committed machine-readable results, $\mathcal{A}$ averages 13.58% of covered pixels. The model trained on crop-noisy labels reads $\kappa = +0.1230$ (sd 0.0361); the matched control, trained on repaired labels and evaluated on identical pixels, reads +0.0157. Their paired difference is $+\mathbf{0.1073}$ (sd 0.0333, $t = 12.88$, positive in 16 of 16 folds, two-sided sign test $p = 0.000031$). Figure 5 shows all machine-readable folds. The power formula returns a value below two, where a paired variance cannot be estimated, so we do not quote a required sample size.

The released text summary contains a rounded seventeenth row, but the corresponding per-fold JSON is absent. Including it would change the paired difference to +0.1106. We exclude that row from every headline result rather than silently mixing machine-readable evidence with an unreconstructable summary.

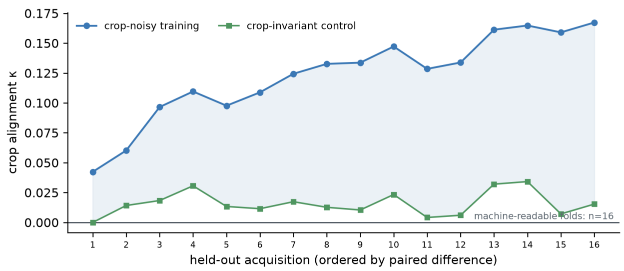

Figure 5. Crop alignment for the 16 held-out acquisitions with machine-readable evidence, each against its matched control. The zero line is the structural null of Proposition 1, not an estimated baseline.

We do not divide this by the calibrated contrast of Section 5 to express it as a fraction of a fully closed-loop labeller. The two were measured on different crop grids, and Section 4.2 shows a grid change moves the contrast by more than any such ratio would report.

Two consistency checks follow. On the repaired label set $\mathcal{A}$ is empty, since crop unanimity of 1.0000 on all 17 folds means every covering crop agrees. And Ω is exactly zero for the two-parameter threshold, because the value channel is a per-pixel function and the thresholds are per-fold constants.

8.3. What the closed loop costs against a human

Sea ice has no reference labels, so it can show that a model learns the labeller but not what that costs. Sen1Floods11 has both. Training on each label set and scoring both against the expert labels on the same held-out event gives a transfer collapse of -0.0454 mIoU (sd 0.0470, $t = -3.21$, negative in 9 of 11 events, two-sided sign test $p = 0.065$). Model input is Sentinel-1 throughout, since the expert labels were produced by analysts correcting a Sentinel-2 classification and the reference arm is open-loop only while the model never reads Sentinel-2.

The mechanism check is harder to explain away. If the collapse comes from learning a labeller rather than a phenomenon, it must be largest where labeller and human disagree most, and it is. The correlation between label agreement and collapse is $+0.719$ with a regression slope at $t = +3.11$. Events where the label sets agree least collapse by -0.0571 against -0.0314 where they agree most, a ratio of $1.8\times$.

9. What these benchmarks can resolve

Throughout we take the acquisitions needed to detect an effect d at 80% power and two-sided $\alpha = 0.05$ as $n = (2.80\, s/d)^2$, with s the paired per-fold standard deviation, $2.80 = z_{0.975} + z_{0.80}$. Each of our effects

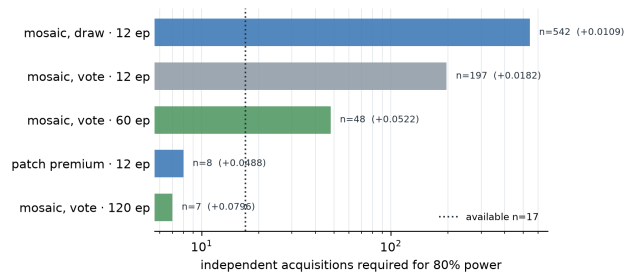

Figure 6. Acquisitions required to detect our artefact-premium estimates at 80% power, log axis. The vertical line marks the 17 available acquisitions. Green bars are longer-budget sensitivity results supported by summary logs rather than per-run JSON and checkpoints.

	premium	sd	t	required n
scoring convention, at the 12-epoch budget				
patch pixels	+0.0488	0.0489	4.12	8
mosaic, vote	+0.0182	0.0910	0.82	197
mosaic, draw	+0.0109	0.0903	0.50	542
training budget, under mosaic-vote scoring				
12 epochs	+0.0182	0.0910	0.82	197
60 epochs	+0.0522	0.1290	1.67	48
120 epochs	**+0.0796**	**0.0747**	**4.39**	**7**

Table 6. The artefact premium under three scoring conventions and, below, at three training budgets on the same 17 folds, with the sample size each would need at 80% power. Budget aggregates come from the committed run log; per-run JSON and checkpoints for the 120-epoch arm are not in the release.

carries its measured paired spread. The normal approximation understates the paired t-test requirement by two to three at these sample sizes, so the values are planning approximations rather than exact thresholds.

We define the artefact premium as a difference in differences: the mIoU gap between a U-Net and a context-free threshold baseline on the closed-loop labels, minus the same gap on the repaired labels, with both arms scored on identical pixels. It is the share of the network's apparent advantage that survives only because of how the labeller was implemented. The same power calculation applies to it.

Patch-pixel scoring needs 8 acquisitions and therefore resolves, but mosaic scoring, which counts each source pixel once, needs 197 and does not. Applying that convention to our own result shrinks the premium $2.7\times$ and removes significance. The draw convention is still more conservative. Estimating rather than fixing the threshold coefficient gives $\hat{\beta} = 0.711$ (se 0.175); under it, the patch-scored premium falls to $t = 1.81$.

Training budget. The committed log reports that the mosaic premium rises from +0.0182 to +0.0796 between 12 and 120 epochs, with the change positive in all 17 folds ($t = 3.98$); required n falls from 197 to 7. The runner records a pre-result stopping criterion: 21 of 34 runs reached the 60-epoch cap without early stopping, compared with 2 of 34 at 120. The 120-epoch result survives every leave-one-out analysis (minimum $t = 4.02$) and four missing-class conventions. These are useful sensitivity checks, but they remain log-level evidence because the release lacks their per-run JSON and checkpoints. We therefore do not use them in the abstract or conclusion.

Multiplicity. This paper reports many tests, and the results it rests on are not close to the boundary. Crop alignment at $t = 12.88$ and the dose-response at $t = 29.84$ would not change interpretation under a Bonferroni factor of 100. The cost at $t = -4.27$ is $p = 0.0016$ on 10 degrees of freedom, so it tolerates a factor of about 30 and not 100, which we state because it is the weakest of the three. One secondary result sits where multiplicity matters and should be read as suggestive: the transfer collapse sign test at $p = 0.065$. We apply no blanket correction, because a structural null and an estimated contrast are not the same kind of object.

10. Recommendations

Generate labels once per scene, preserve nodata as an ignore class, split by acquisition with a receptive-field buffer, and score each source pixel once by mosaicking. Report a context-free baseline, κ and its matched control on a stated crop grid, and the required number of independent units. An empty artefact set means that κ cannot be estimated; the implementation reports this state explicitly, and it must not be read as evidence that the labels are shortcut-free.

11. Limitations

Diagnostic scope. κ requires overlapping crops whose labels disagree. It cannot test a crop-invariant automatic labeller, a scene-level pseudo-label shortcut, or any shortcut unrelated to crop geometry. An empty artefact set is therefore outside the estimand, not a successful negative result.

Identification. Nonzero κ certifies crop-dependent predictions on the artefact set. Even a positive treatment–control contrast does not uniquely prove that the network copied the labelling algorithm: the matched control estimates an architecture and optimisation floor, but label difficulty and training dynamics can still differ between arms. The result is a targeted diagnostic, not a causal mediation analysis.

Scale and external validity. We study two remote-sensing domains, two sensors, and two labelling families. Sea ice contributes 17 acquisitions and floods eleven events. The flood artefact is constructed and is absent from Sen1Floods11 as published; sea ice has no human reference labels. Neither setting establishes how often the failure occurs elsewhere.

Estimand dependence. κ is relative to the chosen crop size, stride, and weighting, so its magnitude does not travel between papers. It also does not directly quantify score inflation: our unfitted prediction $P(\mathcal{A})\kappa$ gets the sign right on 14 of 17 folds but misses the magnitude by a factor of 1.6–1.9 and has fold-level rank correlation +0.216.

Evidence boundary. The machine-readable sea-ice summary contains 16 paired fold values, while a rounded text report contains an unmatched seventeenth. The flood dial and accuracy-cost event rows and the 60/120-epoch aggregates are also committed as summary logs. Raw predictions, checkpoints, and some per-run JSON are absent. The release therefore verifies the arithmetic of the supported aggregates; it does not reproduce training or regenerate every result end to end. We exclude the unmatched fold from the main result and keep longer-budget results as sensitivity evidence. The supplement inventories these tiers and the claims withdrawn from earlier drafts.

12. Conclusion

Closed-loop labelling is usually discussed as a kind of noise. It is the opposite of noise, because a model can learn it, and that is why it inflates measured performance rather than depressing it. We gave a test whose null follows from the definition rather than from a control arm, verified it against a case with a known answer, calibrated it on a labeller we could turn up and down, and showed that what it detects costs accuracy against human annotation. On the benchmark where the artefact occurred naturally the paired contrast is +0.1073 across all 16 folds with machine-readable evidence. At the initial mosaic-scored budget, however, the aggregate artefact premium itself is unresolved (required $n = 197$ versus 17 available acquisitions), and the stronger longer-budget result is summary-log evidence only. Crop alignment makes one specific shortcut measurable; the provenance audit defines which claims the current release can support.

Code and data. Code and auditable artifacts are included in the anonymous supplementary submission. The public URL is withheld during double-blind review and is enabled in the camera-ready source.

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